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Welcome





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The Story Behind the Challenge

Robots cannot be programmed for every object, variation and situation – they need a repeatable learning pipeline.

THE BACKGROUND

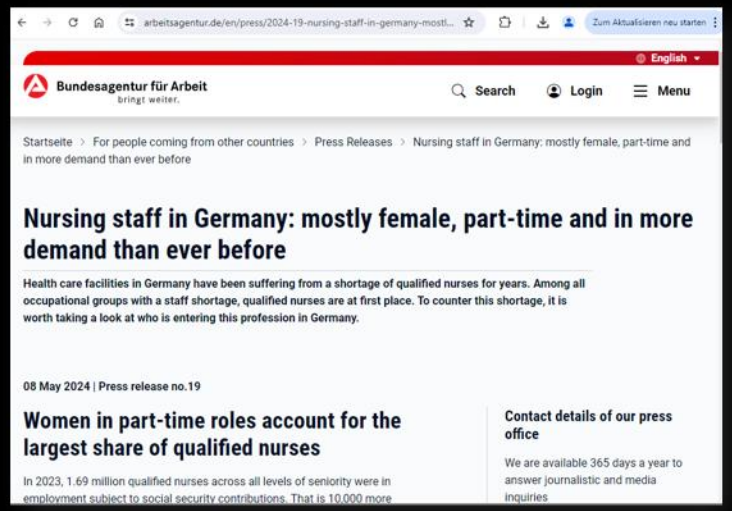
Cardiac Catheter Lab Nurse

Responsible for:

- patient preparation and monitoring
- equipment preparation
- assist cardiologists

The impulse:

- acute staff shortage
- lack in experience at staff



Bundesagentur für Arbeit

bringt weiter.

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Startseite › For people coming from other countries › Press Releases › Nursing staff in Germany: mostly female, part-time and in more demand than ever before

Nursing staff in Germany: mostly female, part-time and in more demand than ever before

Health care facilities in Germany have been suffering from a shortage of qualified nurses for years. Among all occupational groups with a staff shortage, qualified nurses are at first place. To counter this shortage, it is worth taking a look at who is entering this profession in Germany.

08 May 2024 | Press release no.19

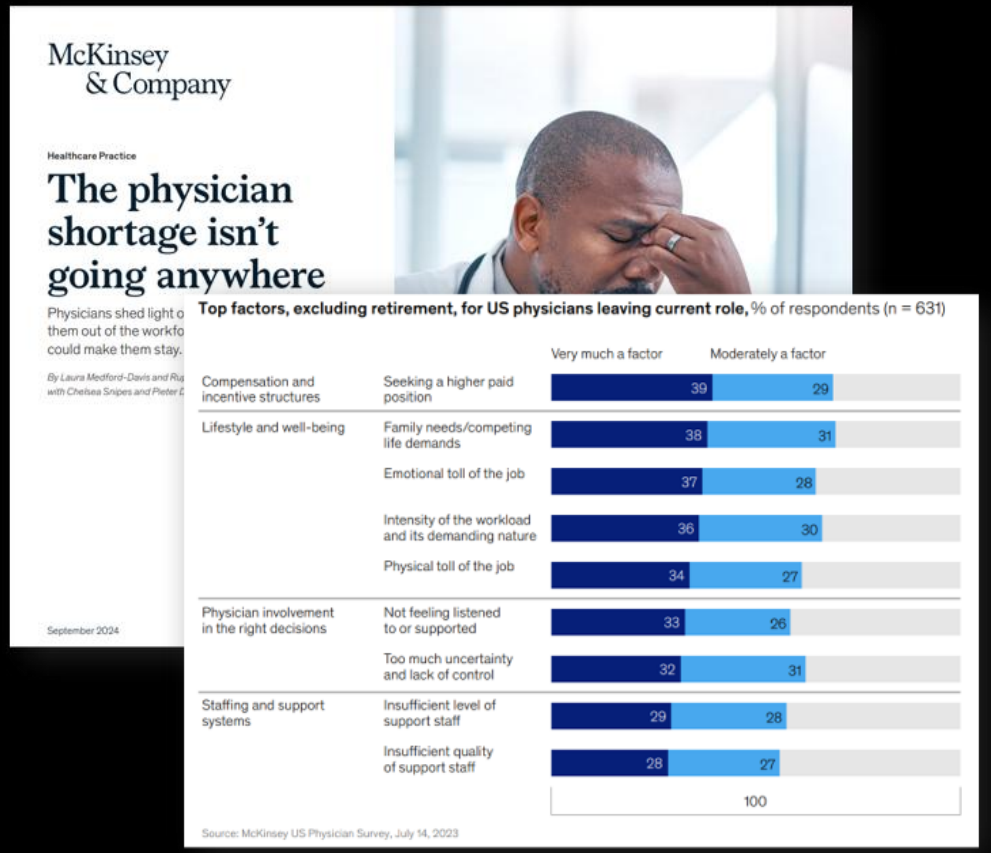
Women in part-time roles account for the largest share of qualified nurses

In 2023, 1.69 million qualified nurses across all levels of seniority were in employment subject to social security contributions. That is 10,000 more

Contact details of our press office

We are available 365 days a year to answer journalistic and media inquiries

E.g., the lack of nurses is driving the healthcare system to its limit in Germany..



McKinsey & Company

Healthcare Practice

The physician shortage isn't going anywhere

Physicians shed light o
them out of the workfo
could make them stay.

By Laura Medford-Davis and Ru
with Chelsea Snipes and Pieter E

September 2024

Top factors, excluding retirement, for US physicians leaving current role, % of respondents (n = 631)

		Very much a factor	Moderately a factor
Compensation and incentive structures	Seeking a higher paid position	39	29
Lifestyle and well-being	Family needs/competing life demands	38	31
	Emotional toll of the job	37	28
	Intensity of the workload and its demanding nature	36	30
	Physical toll of the job	34	27
Physician involvement in the right decisions	Not feeling listened to or supported	33	26
	Too much uncertainty and lack of control	32	31
Staffing and support systems	Insufficient level of support staff	29	28
	Insufficient quality of support staff	28	27

100

Source: McKinsey US Physician Survey, July 14, 2023

..and pushing physicians to quit their positions in the US..

Matthias Hoff | SHS TC MP TI, Luca Hilling | SHS IT GD&AI IPA

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The Story Behind the Challenge

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Robots cannot be programmed for every object, variation and situation – they need a repeatable learning pipeline.

THE BACKGROUND

Cardiac Catheter Lab Nurse

The idea:
robotic assistance in cathlabs

TODAY

Traditional Robot Automation

Task-specific programming

Fixed conditions

High adaptation effort

FROM PROGRAMMING

WHAT WE WANT

Physical AI

Learning from demonstrations

Adaptation to variable objects and scenes

Reusable AI-driven capabilities

TO DEMONSTRATING AND LEARNING







We do not only want a robot that performs one task. We want the learning loop that creates, evaluates and deploys new robot capabilities.

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Core Challenge – Teach the Robot

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Build the Autonomous Learning Loop

01
Ingest & Transform

How can we turn raw robot demonstrations into reliable training data?
How do we preserve the correct observations, actions and episode structure? Should data be enriched?

02
Dataset Intelligence

How can we understand whether the dataset is ready for training?
Which episodes are invalid, redundant or unrepresentative, and what is still missing?
Is the dataset ready for VLAs?

03
Fine-Tune

How can we turn the demonstrations into an adaptable robot capability?
Which model, training strategy and configuration are most suitable for the task?

04
Evaluate

How can we measure what the robot has learned and where it still fails?
How can progress, failure stages, robustness and latency be assessed reproducibly?

05
Version & Serve

How can we make the resulting capability reproducible, accessible and ready to use?
How can models be connected to the virtual or real world? What is the best deployment and serving strategy?

STAGE 5 FEEDS THE NEXT LEARNING CYCLE

Raw demonstrations in. A served, evaluated and reproducible robot model out.



SCOPE

Simulation-first – virtual environment is provided
Focus on object manipulation
Prepared demonstrations are provided
Navigation is not part of the mandatory core

DEFINITION OF DONE

Fine-tuned model checkpoint & demonstratable physical AI stack from 01 to 05
Team presents task execution with their own expert policy
Rich concepts can sometimes be more valuable than poor results.

Luca Hilling | SHS IT GD&AI IPA
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Go Beyond the Core

Choose an innovation direction or invent your own

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Core Model

+ + + + +

OPTION
A
Intelligent Success Judgement
Assess rollouts with vision, VLMs or state traces. Explain if, how well and where execution degraded.

OPTION
B
Self-Improving Pipeline
Find capability gaps, recommend missing data and propose the next training strategy.

OPTION
C
Learning Network
Share skills, experience and validated model versions across robots, embodiments or sites.

OPTION
D
Skill Orchestration
Decompose multi-step requests, pick the right policy per subtask, add safety and latency guardrails.

OPTION
X
Innovation Joker
Propose an original extension that adds value for the clinical scenario or the learning loop.

→ The best idea may be the one we have not thought of yet.

From first concept to working prototype, every level of exploration is welcome. Show us your idea, your reasoning and how you would take it further.

Complete the mandatory core first. Then choose one or more optional modules for demonstration.

Luca Hilling | SHS IT GD&AI IPA
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What we provide

You will get a dataset with 500 episodes, the full 3D clinical environment with the FR3 Duo and detailed information

DATA

Dataset

Team-specific dataset on the local file share. Download it with your link, QR code and key, then upload it to your team workspace.



SIMULATION

3D Environment

Team-specific 3D environment on the local file share. Download it with your link, QR code and key, then upload it to your team workspace.



DOCS

Detailed Instructions

Setup, data schema and evaluation rules will be available in the same local file-share web app.

QR

[DOCS LINK]

PEOPLE

Contacts & Mentoring

Challenge owners and mentors on site, plus a support channel during the hackathon.



Luca Hilling
Challenge Owner and Mentor
On site



Matthias Hoff
Challenge Owner and Mentor
On site



Arjen Heimstädt
Mentor
On site



Thomas Mutzke
Mentor
On site



Thomas Bursian
Challenge Owner and Mentor
Remote / CN time zone

FURTHER RESOURCES

Toolchain / Stack

- Microsoft Physical AI Toolchain
- Other?

Vision Language Action Models

- Physical Intelligence openpi repository
- Physical Intelligence
- NVIDIA GR00T

Virtual Environment & Simulation

- NVIDIA Isaac Sim
- MuJoCo

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Where we want to go

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Our real Robot – Franka Robotics “FR3 Duo”



The virtual world



Synthetic Data through virtual Environments



Real VLA Model inference







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Assets

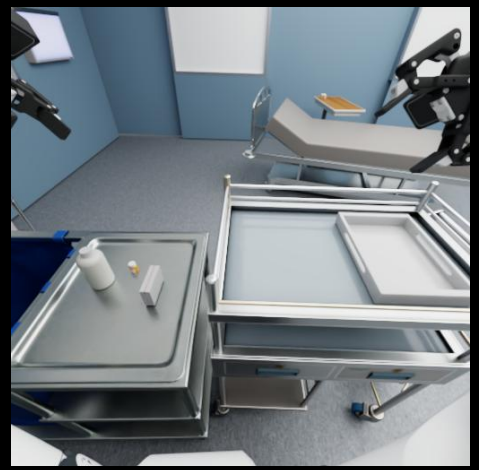
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Physical AI for adaptive robotic assistance in a clinical environment













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